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When is a constant rate truly constant?

A Monte Carlo power analysis of the logistic operationalization of Constant Rate Effects

RUSE 2019

20–21 August 2019

MANCHESTER 1824
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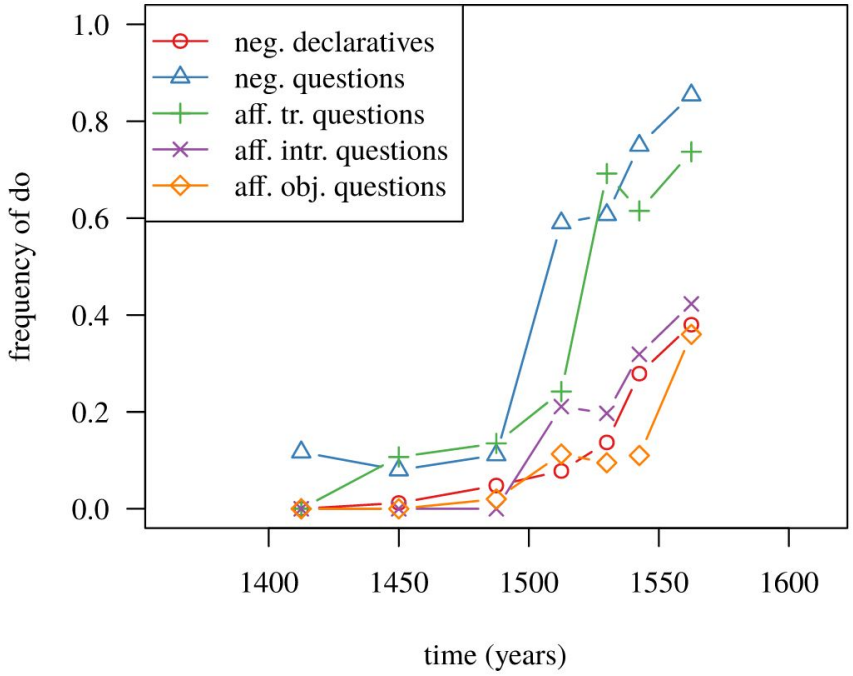
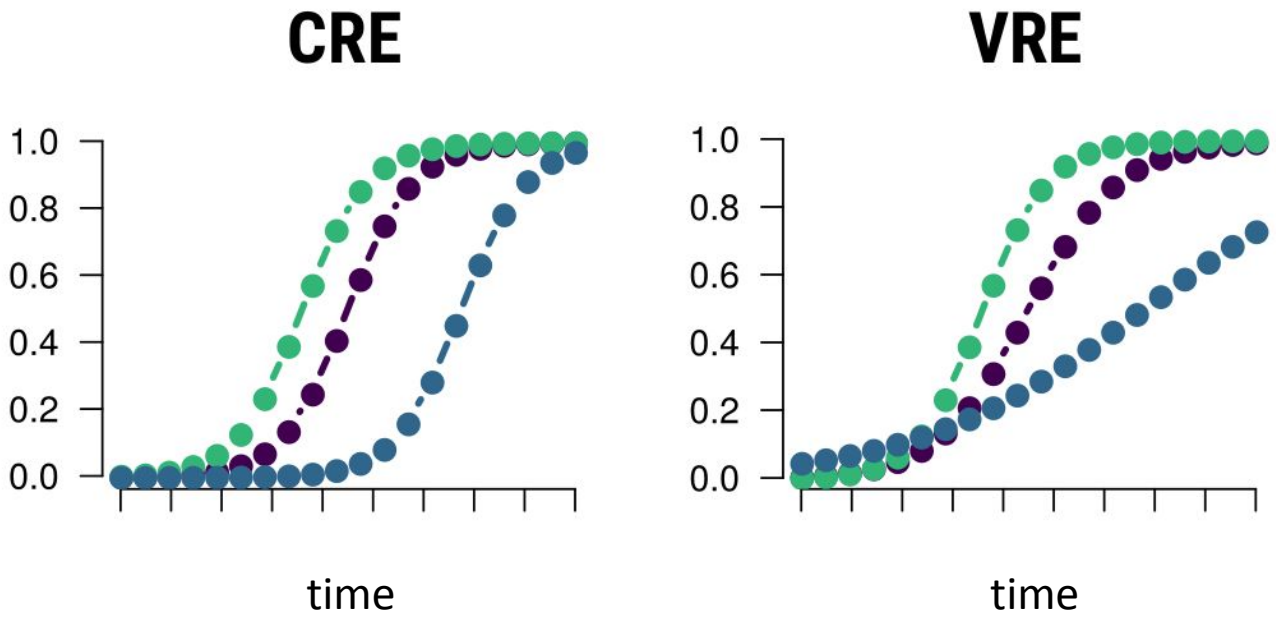
Constant Rate Hypothesis (CRH)

"[W]hen one grammatical option replaces another with which it is in competition across a set of linguistic contexts, the rate of replacement, properly measured, is the same in all of them."

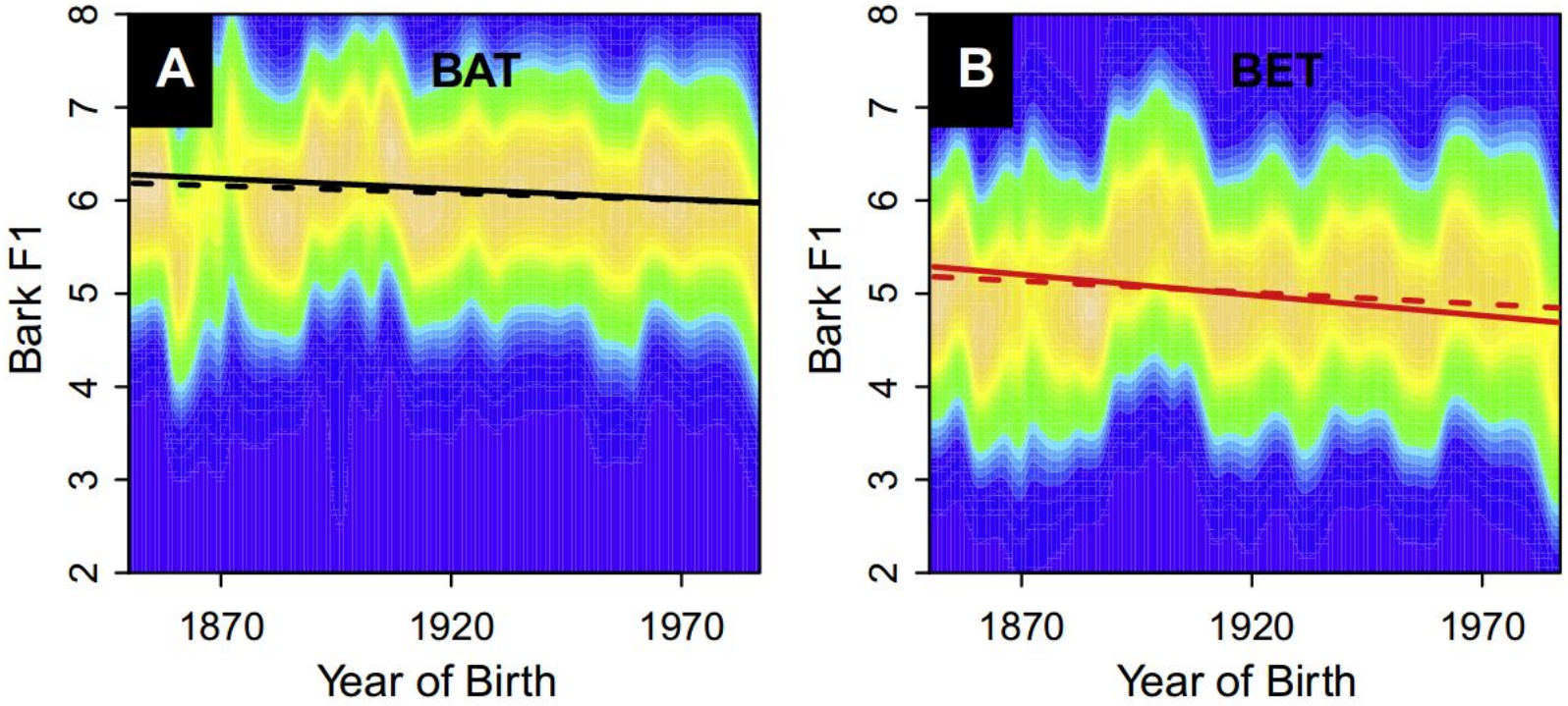
(Kroch 1989b:200)



Constant vs Variable Rate Effects



N.B. Classically the variable is categorical (0 vs 1) – points on the curve represent population averages.



Hay, Pierrehumbert, Walker & LaShell (2015)

The classical (logistic) operationalization

Assume changes are S-shaped (logistic function)

Two models: VRE (slopes can differ), CRE (slopes identical)

Two related methods in the literature:

1. Likelihood ratio test (Kroch 1989a, 1989b, many others)
2. Logistic regression (e.g. Fruehwald, Gress-Wright & Wallenberg 2013)

CRE is cast as the null hypothesis!

Paolillo's Problem

- This results in a problem pointed out by Paolillo (2011):

  Every CRE decision carries an unknown probability of a type II error
  (since we report a CRE whenever we *fail to show that it is a VRE*)

- We can have no confidence in a CRE claim unless the method's power is known
- Power π relates to type II error β as: $\pi = 1 - \beta$
- Power of the logistic operationalization of CREs is unknown
  - One notable exception: Ecay (2015) conducts a power analysis of Kroch (1989b)

An *a priori* Monte Carlo power analysis

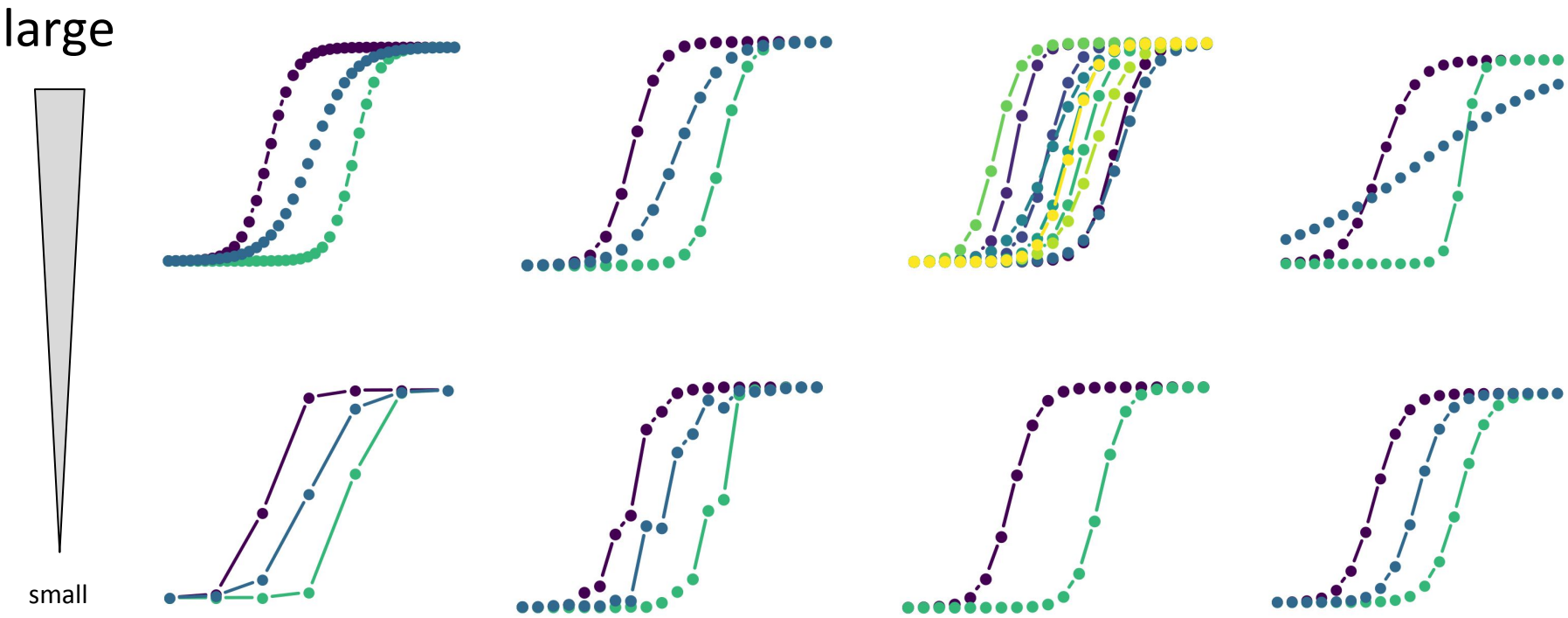
- Take the two models, CRE and VRE.
- Generate **synthetic data** from the VRE model.
- Apply the classical method (logistic regression) to get a decision: CRE/VRE.
  - Decision is CRE if time:context interaction term is significant at the $p < 0.05$ level
- Repeat many, many times.
- Obtain proportion of experiments where decision is incorrect (i.e. decision is CRE): this is the type II error rate β
- Repeat for various different **data resolution** parameters and **effect sizes** to get an idea of how these affect β.

“horizontal” resolution:
number of times *T*

“vertical” resolution:
sampling resolution *S*

“lateral” resolution:
number of contexts *N*

effect size:
magnitude of VRE



horizontal resolution:
number of times *T*

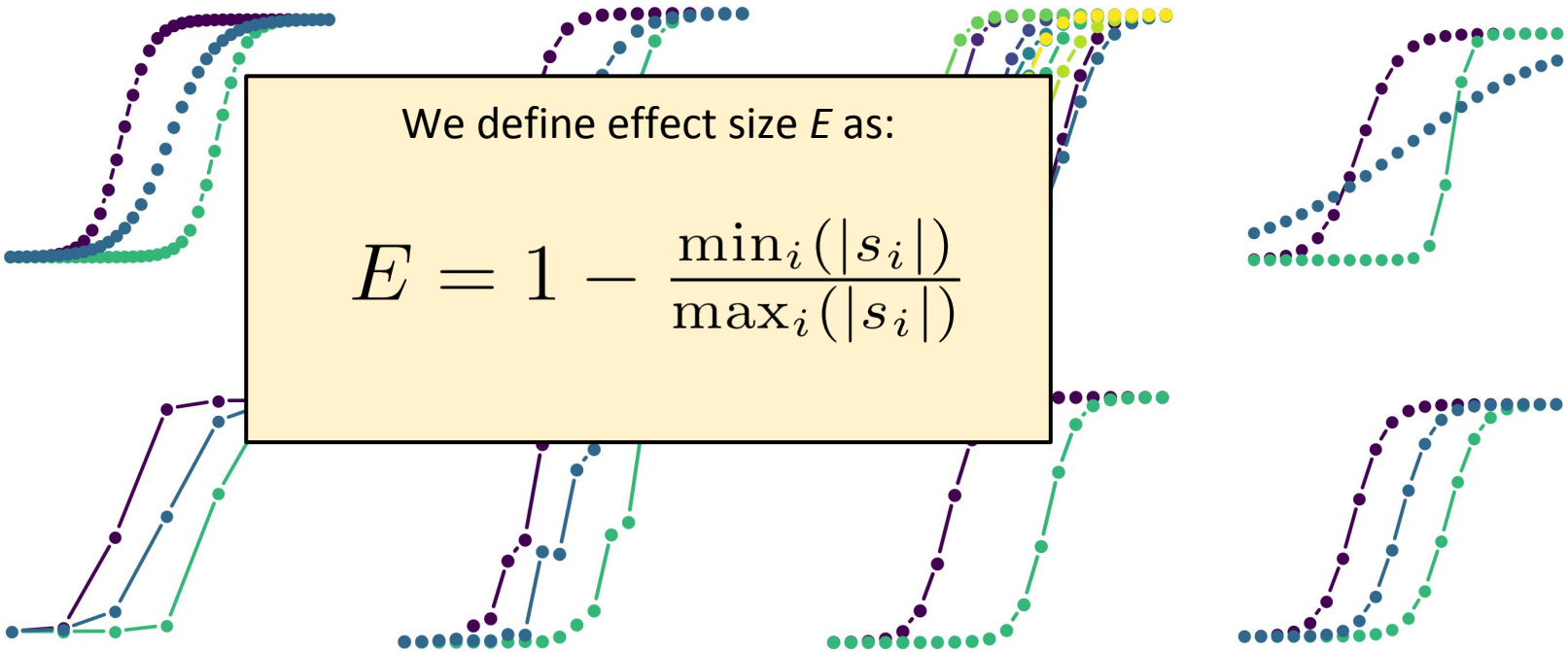
vertical resolution:
sampling resolution *S*

lateral resolution:
number of contexts *N*

effect size:
magnitude of VRE

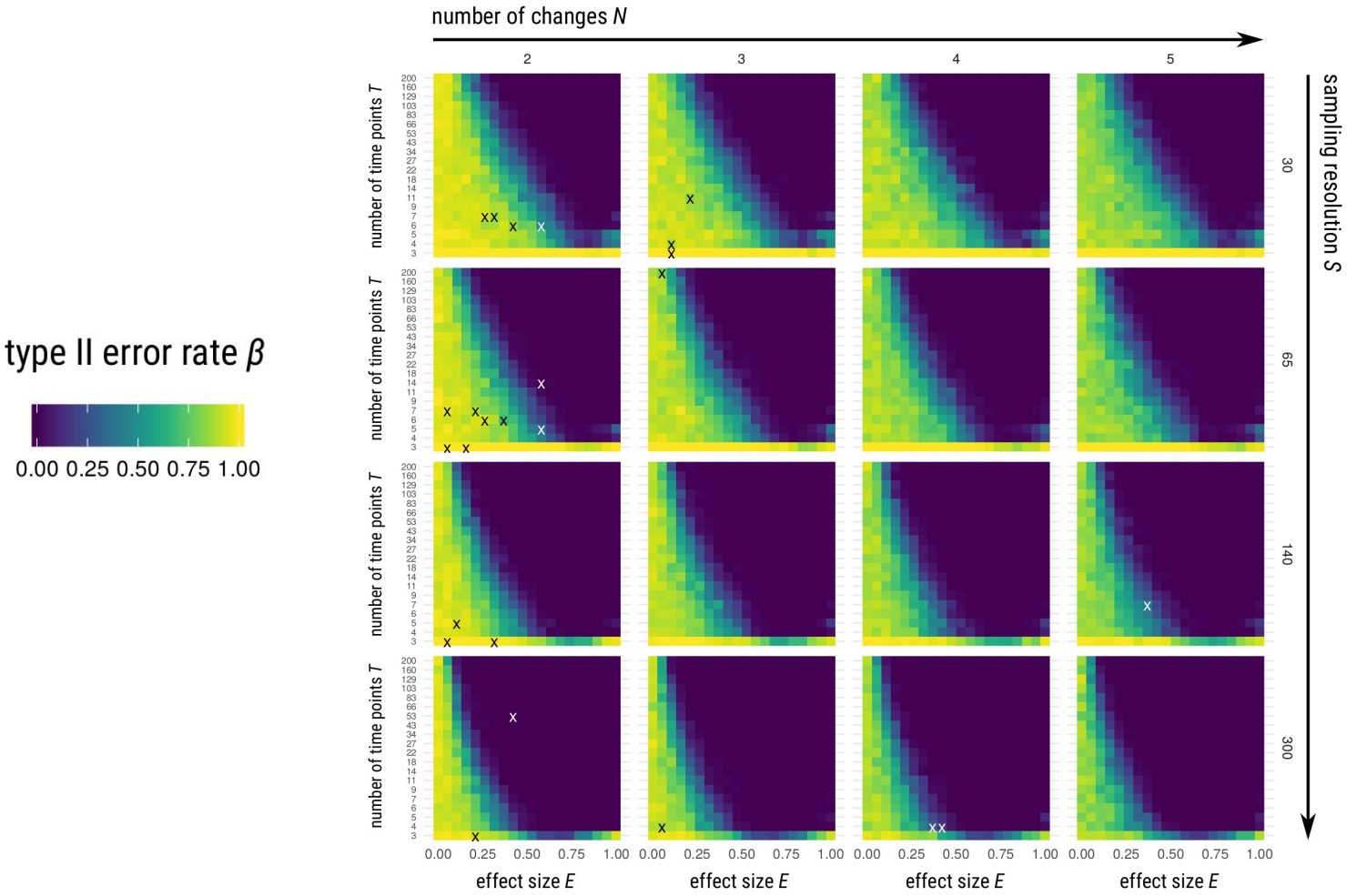
large

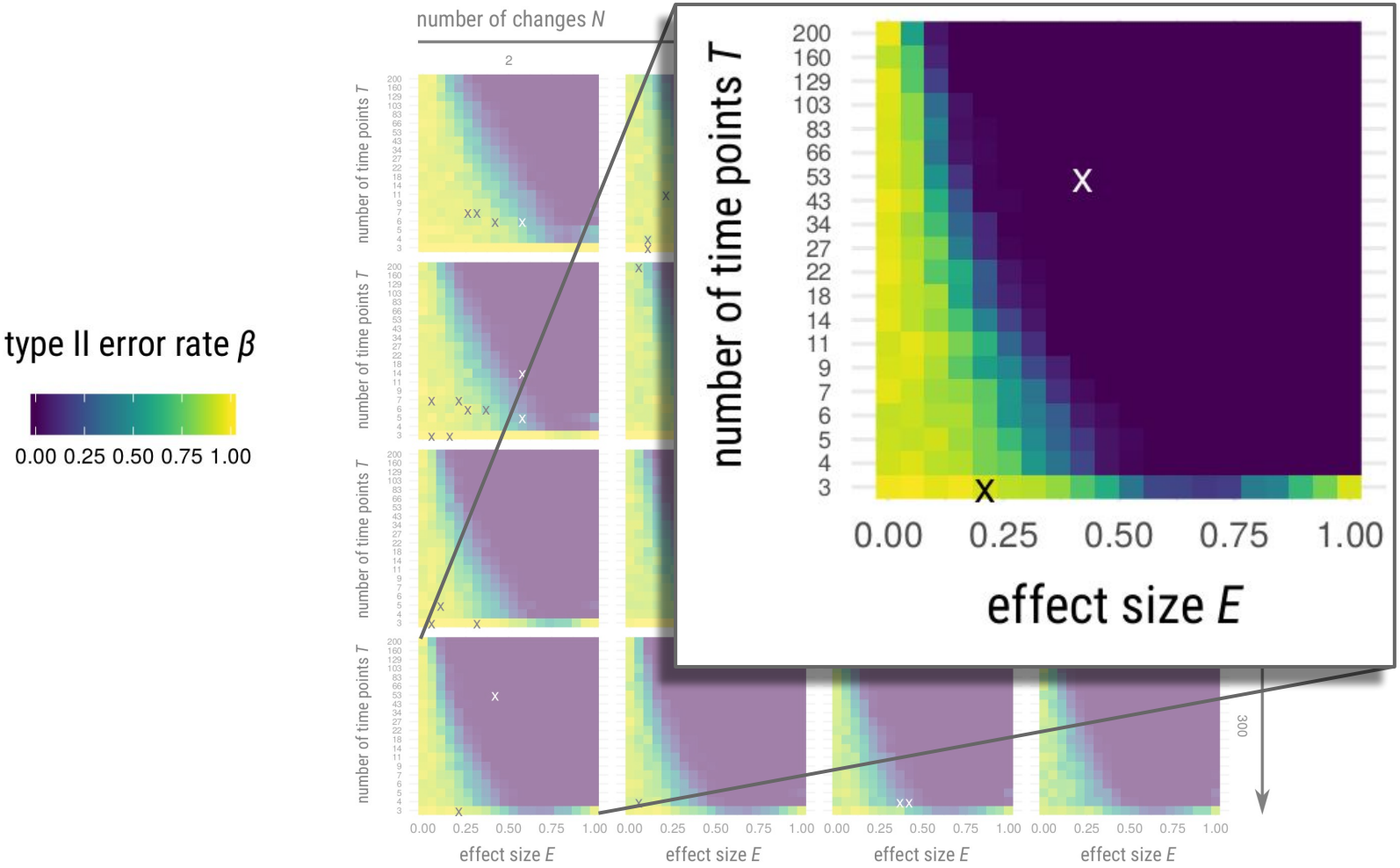
small



We define effect size *E* as:

$$E = 1 - \frac{\min_i(|s_i|)}{\max_i(|s_i|)}$$


number of changes N
2
3
4
5
sampling resolution S
30
65
140
300
number of time points T
effect size E
0.00 0.25 0.50 0.75 1.00
type II error rate β
0.00 0.25 0.50 0.75 1.00


type II error rate β
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number of changes N
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number of time points T
effect size E
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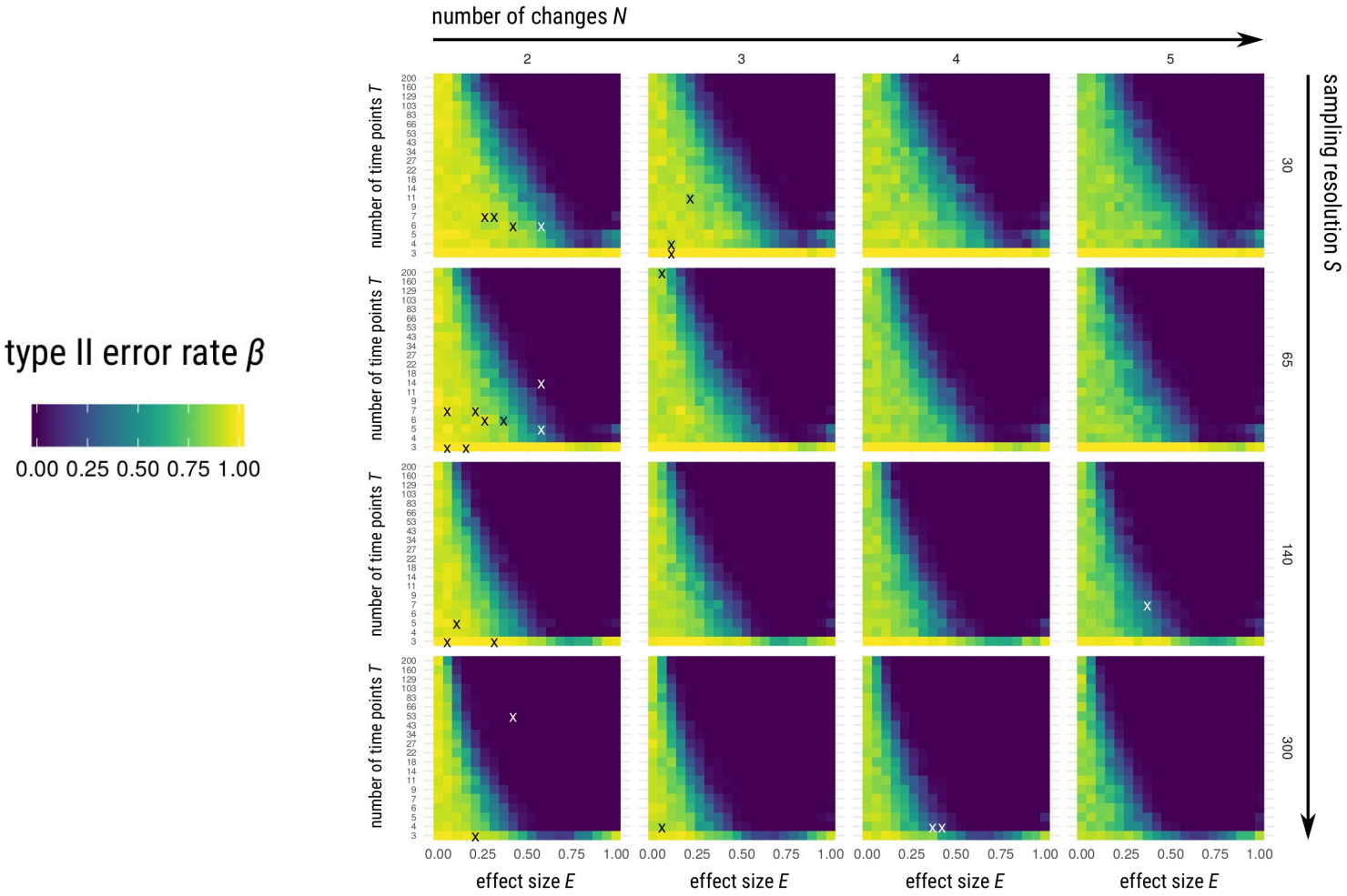
number of changes N
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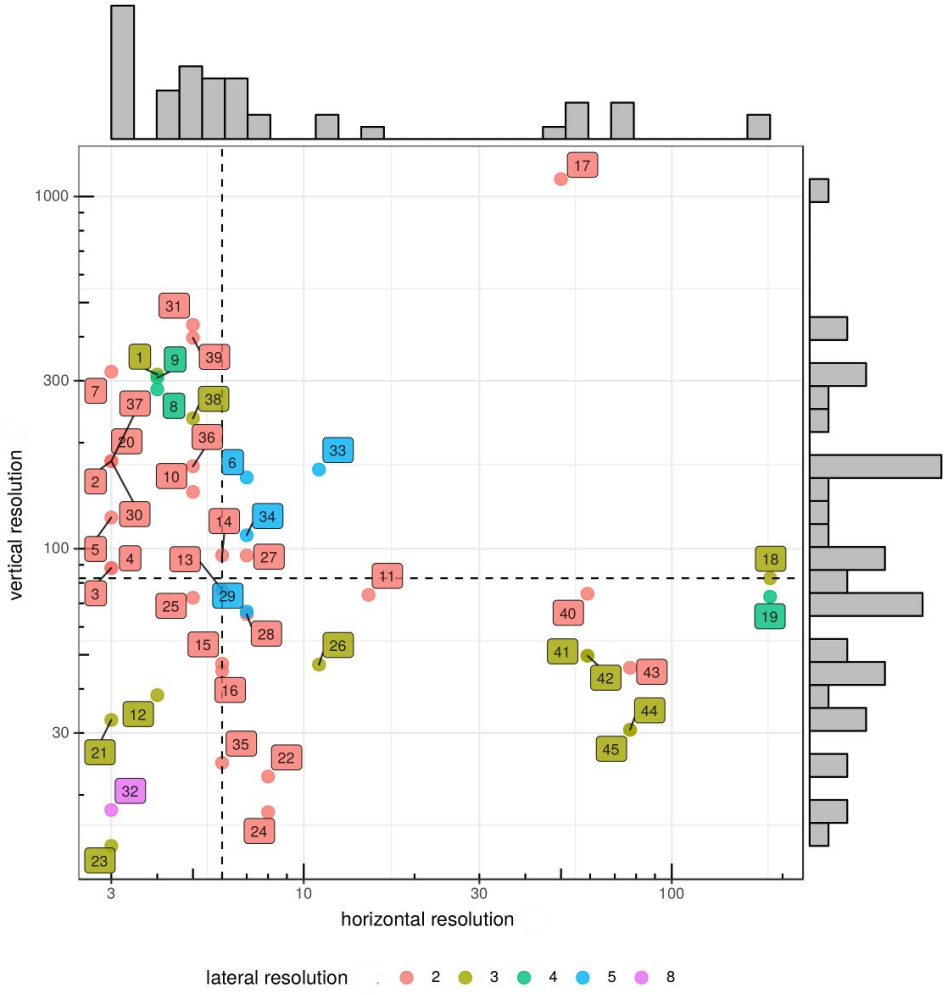
300
number of time points T
effect size E
0.00 0.25 0.50 0.75 1.00
type II error rate β
0.00 0.25 0.50 0.75 1.00

So what are those little X's?

We've compiled a database (the **Constant Ratabase**) of changes from existing literature:

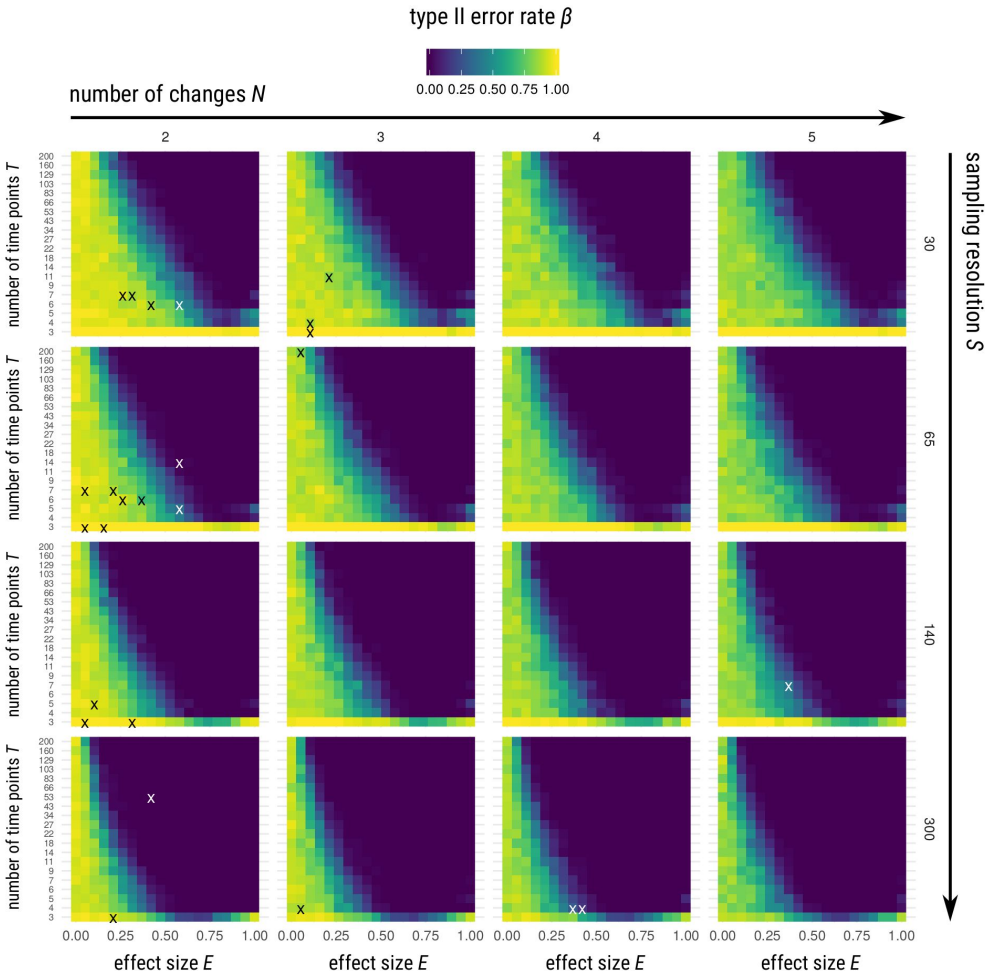
- 45 changes in total (in 8 languages)
  - 27 putative CREs
  - 10 putative VREs
  - 8 new datasets not previously probed for a CRE
- Publications range from 1989 to "in press"
- Changes range from Middle Ages to present day

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	We hav	Use it	File	ID	Dependent variable	Type	C	T	Publication(s)	Data from	Original method	Method	Sigmoidal	Dependent variab	Conte
2	TRUE	TRUE	kroch_do.cs		English do-support (fu	VRE			Kroch (1989)	Table 3	χ^2 on logistic slop	TRUE	FALSE	English *do*-suppor	Affirm
3	TRUE	TRUE	kroch_do_li		English do-support (fin	CRE			Kroch (1989)	Table 3	χ^2 on logistic slop	TRUE	TRUE		
4	FALSE	TRUE			French V2				Kroch (1989)					French V2	Null s
5	FALSE	TRUE			Portuguese possessives and definit				Kroch (1989)					Portuguese poss	Uniqu
6	TRUE	TRUE	kroch_have		British English have v	CRE			Kroch (1989)	Table 1	VARBRUL	TRUE		British English *ha*	Tempo
7	TRUE	TRUE	kroch_have		British English have v	CRE			Kroch (1989)	Table 2	VARBRUL	TRUE		British English *ha*	Concre
8	TRUE	TRUE	wallage_1v		English Jespersen Cyc	CRE			Wallage (2013)	Table 1	Eyeballing of fact	FALSE	TRUE	English Jespersen	Discou
9	TRUE	TRUE	wallage_1v		English Jespersen Cyc	CRE			Wallage (2008)	Tables 8 & 9	Eyeballing of fact	FALSE		English Jespersen	Main
10	TRUE	TRUE	wallage_2v		English Jespersen Cyc	CRE			Wallage (2008)	Tables 11 & 12	Eyeballing of fact	FALSE		English Jespersen	Main
11	TRUE	TRUE	german_fo		German final fortition	CRE			Fruehwald et al. (2	Table 1	Regression intera	TRUE	TRUE	German final fort	/b/, /c
12	TRUE	FALSE	german_fo		German final fortition	CRE			Fruehwald et al. (2	Joel Wallenberg (p	Regression intera	TRUE	TRUE	German final fort	/b/, /c
13	TRUE	FALSE	german_fo		German final fortition	CRE			Fruehwald et al. (2	Joel Wallenberg (p	Regression intera	TRUE	TRUE	German final fort	/b/, /c
14	TRUE	FALSE	german_fo		German final fortition	CRE			Fruehwald et al. (2	Joel Wallenberg (p	Regression intera	TRUE	TRUE	German final fort	/b/, /c
15	TRUE	FALSE	german_fo		German final fortition	CRE			Fruehwald et al. (2	Joel Wallenberg (p	Regression intera	TRUE	TRUE	German final fort	/b/, /c
16	TRUE	FALSE	german_fo		German final fortition	CRE			Fruehwald et al. (2	Joel Wallenberg (p	Regression intera	TRUE	TRUE	German final fort	/b/, /c
17	TRUE	FALSE	german_fo		German final fortition	CRE			Fruehwald et al. (2	Joel Wallenberg (p	Eyeballing of traj	FALSE	FALSE		
18	TRUE	FALSE	german_fo		German final fortition	?			Fruehwald et al. (2	Joel Wallenberg (p	N/A	FALSE	FALSE		
19	TRUE	FALSE	german_fo		German final fortition	?			Fruehwald et al. (2	Joel Wallenberg (p	N/A	FALSE	FALSE		
20	TRUE	FALSE	german_fo		German final fortition	?			Fruehwald et al. (2	Joel Wallenberg (p	N/A	FALSE	FALSE		
21	TRUE	FALSE	german_fo		German final fortition	CRE			Fruehwald et al. (2	Joel Wallenberg (p	Eyeballing of traj	FALSE	FALSE		
22	TRUE	TRUE	santorini_u		Yiddish INFL headedn	CRE			Santorini (1993)	Table 1	χ^2 on logistic slop	TRUE		Yiddish INFL head	Simpl
23	TRUE	TRUE	santorini_a		Yiddish INFL headedn	CRE			Santorini (1993)	Table 6 & conflate	χ^2 on logistic slop	TRUE		Yiddish INFL head	Simpl
24	FALSE	TRUE			Greek VP headedness				Taylor (1994)					Greek VP headed	Differ
25	TRUE	TRUE	pintzuk.csv		English VP headedne	CRE			Pintzuk (1995)	Table 3	χ^2 on logistic slop	TRUE		English VP heade	Main
26	FALSE	TRUE			Welsh pronoun chdi				Willis (2017)					Welsh second-per	Differ
27	TRUE	FALSE	rc_extrapo		Relative clause extrap	VRE			Wallenberg (2016)	Joel Wallenberg (p.c.)				Relative clause ex	Englis
28	TRUE	TRUE	frisch_2vs3		English not as senten	CRE			Frisch (1997)	Table 5	χ^2 on model fit	TRUE		Change of *not* fro	With
29	TRUE	FALSE	frisch_1vs3		Parameter linking ne	VRE			Frisch (1997)	Table 14	χ^2 on model fit	TRUE		English Jespersen	*ne* los
30	TRUE	TRUE	frisch_1vs3		Parameter linking ne	VRE			Frisch (1997)	Table 14	χ^2 on model fit	TRUE		English Jespersen	*ne* los
31	TRUE	TRUE	sundquist_n		Parameter linking en	VRE			Sundquist (2007)	Table 1, without Ty	Eyeballing of slop	FALSE		Middle Low Germ	en los
32	TRUE	TRUE	springville_		Be like vs. other quot	CRE			Cukor-Avila (2002)	Table 6	VARBRUL	TRUE		*Be like* vs. other q	First v
33	TRUE	TRUE	springville_		Be like vs. other quot	VRE			Cukor-Avila (2002)	Table 7	VARBRUL	TRUE		*Be like* vs. other q	Dialog
34	TRUE	TRUE	springville_		Say vs. other quotativ	CRE			Cukor-Avila (2002)	Table 9	VARBRUL	TRUE		*Say* vs. other quot	First v
35	TRUE	TRUE	springville_		Say vs. other quotativ	VRE			Cukor-Avila (2002)	Table 10	VARBRUL	TRUE		*Say* vs. other quot	Dialog
36	TRUE	TRUE	kallel_cooro		Negative concord in E	CRE			Kallel (2005, 2007)	2007, Tables 1 & 2	Eyeballing of slop	FALSE		Negative concord	Co-ord
37	TRUE	TRUE	kallel_obj.c		Negative concord in E	CRE			Kallel (2005, 2007)	2007, Tables 3 & 4	Eyeballing of slop	FALSE		Negative concord	Object
38	TRUE	TRUE	kallel_adv.c		Negative concord in E	CRE			Kallel (2005, 2007)	2007, Tables 5 & 6	Eyeballing of slop	FALSE		Negative concord	Adverl
39	TRUE	FALSE	durham_et		Be like vs. other quot	CRE			Durham et al. (20	Daniel Ezra Johnso	Regression intera	TRUE		*Be like* vs. *say* in t	For-ac
40	TRUE	FALSE	pintzuk_tay		English VP headedne	FIXME			Pintzuk & Taylor (2	Table 11.9	FIXME			English VP heade	Positiv
41	TRUE	TRUE	egyptian.cs		Egyptian possessive c	CRE			Gardiner (2015)	Table 8	Eyeballing of logi	FALSE	TRUE	Egyptian possessi	Officia
42	TRUE	TRUE	french_subj		Null subjects in Frenc	CRE			Simonenko et al. (2	Alexandra Simone	N/A	FALSE	TRUE	Null subjects in th	Main
43	TRUE	TRUE	sundquist_p		XV vs. VX in Norwegi	CRE			Sundquist (2002, 2	2006, Table 2	Eyeballing of logi	FALSE		XV vs VX in Norw	DP obj
44	TRUE	TRUE	tagliamonte		Be like vs. other quot	?			See Appendix	Sali Tagliamonte (p	N/A	FALSE		be like vs. other q	singul
45	TRUE	TRUE	tagliamonte		Be like vs. other quot	?			See Appendix	Sali Tagliamonte (p	N/A	FALSE		be like vs. other q	past, p
46	TRUE	TRUE	tagliamonte		Be like vs. other quot	?			See Appendix	Sali Tagliamonte (p	N/A	FALSE		be like vs. other q	first, s


vertical resolution
1000
300
100
30
horizontal resolution
3
10
30
100
lateral resolution
2
3
4
5
8

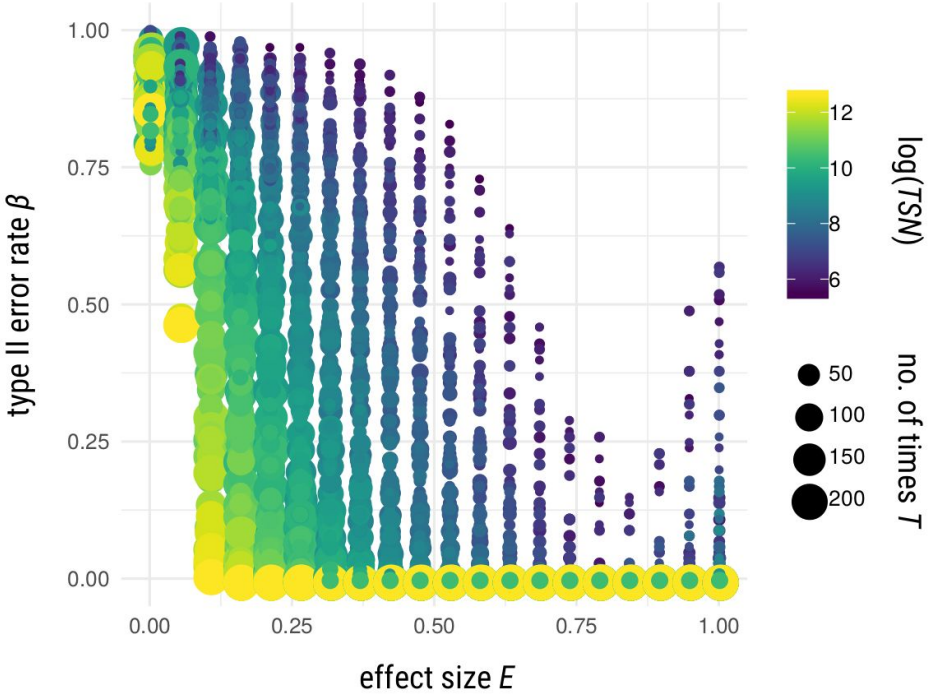
Results

- Of the 27 putative CREs, method fails to converge on 2
- Of the remaining 25, **only 3** reach a power of 0.8 (conventional threshold)
- Several studies have a horizontal resolution of $T = 3$ time points and a vertical resolution $S < 100$
- But the method is useless on such datasets! (Type II error rate is always > 0.2 and often as high as 1.0.)
- **The classical operationalization cannot be used on traditional small datasets.**



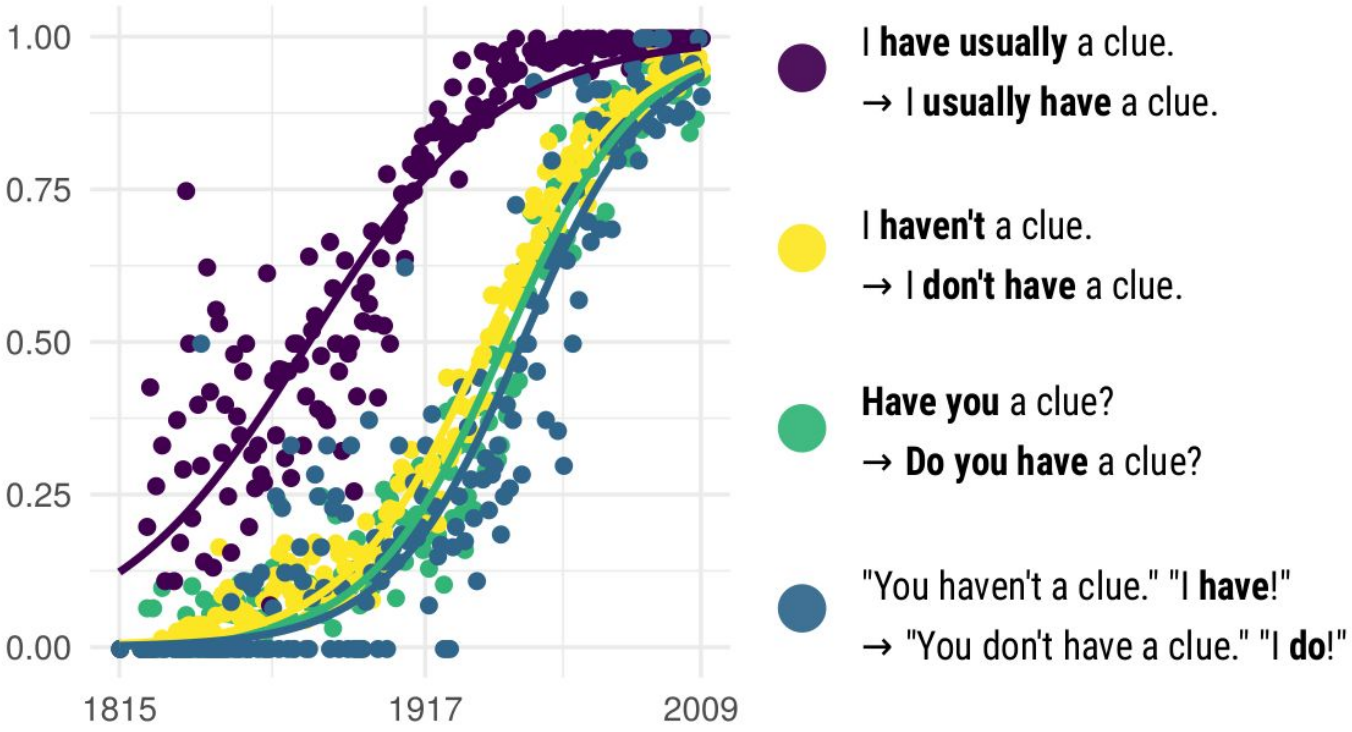
Main take-homes

- Sufficient data resolution is **crucial** – otherwise a CRE claim is (at least statistically) meaningless
- Datasets with low horizontal resolution represent the most difficult case
- In practical terms – to guide and inform applied research – we need
    - a formula that gives estimated power as a function of the data resolution parameters, or
    - a software package that calculates the (exact a priori) power given those parameters
- We're working on both of these – stay tuned!



What is to be done?

1. More (and better) data
2. Alternative models



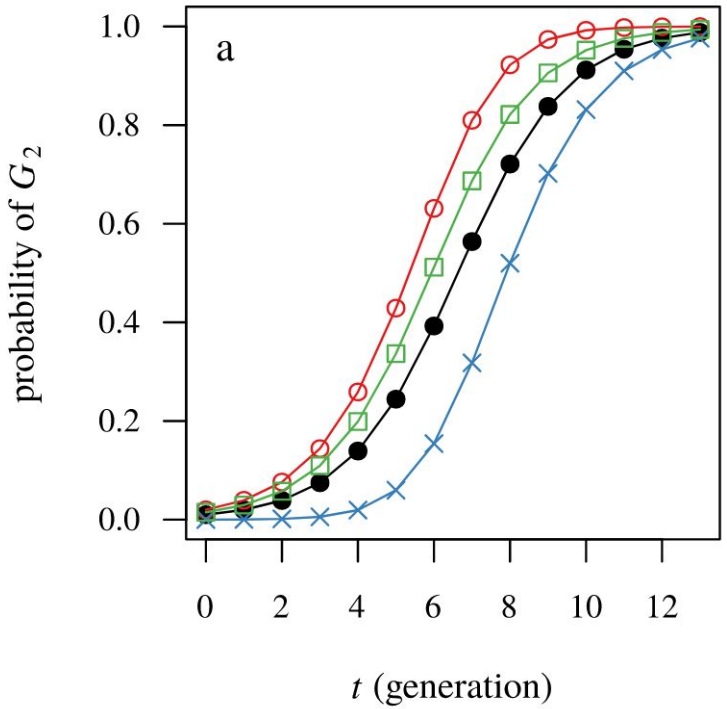
Data from Zimmermann (2017)

Bias model

$$\begin{cases} q_t^{(1)} = \tilde{q}_t + b_1 \tilde{q}_t (1 - \tilde{q}_t) \\ \vdots \\ q_t^{(K)} = \tilde{q}_t + b_K \tilde{q}_t (1 - \tilde{q}_t) \end{cases}$$

Kauhanen & Walkden (2018):

- CRE is derived from one underlying (logistic) curve by a production/perception bias mechanism
- Each context curve is almost but not quite logistic
- Will this mean model is easier to tell apart from a VRE than the classical CRE model?



Moreover…

- What to do with non-logistic, possibly non-monotonic changes?
- Kauhanen & Walkden's (2018) model extends to such cases in theory, but no effective curve-fitting algorithm is currently known…
- And what would the VRE model be in this case?
- The changes we've looked at so far all deal with categorical variables. Can the framework be extended for continuous variables, too?
- Interesting questions for future research!

Envoi: A bit of philosophy

- The slopes in a VRE may be (near) identical for accidental reasons
  - Wallenberg (2016): a “CRE” across languages
  - Willis (2017): a “CRE” across geographical regions
- This means that statistics alone is insufficient to decide between CREs and VREs
- For any effect size (magnitude of VRE) E, some data resolution vector (T, S, N) exists that takes the type II error β below any desired threshold
- This means that an opponent of the CRH may always claim: “You simply have not collected enough data to detect the variable rate”
- **So what is the role of sophisticated statistical methods in this type of research?**

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